



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CSE
Course Outcome	CO - 1

Case Study topic: UrbanAI – Intelligent Smart City Infrastructure Optimization Platform

Work Done Summary:

In this case study, we implemented UrbanAI, an Intelligent Smart City Infrastructure Optimization Platform. The platform uses a self-balancing tree structure to manage smart city service records including traffic nodes, utilities, and public services in sorted order. The system automatically maintains balance after every insertion, ensuring $O(\log n)$ access times for all city infrastructure queries. We implemented multiple city zone registrations and observed automatic rebalancing operations to maintain optimal query performance.

The UrbanAI Tree stores city zone titles in sorted order and automatically maintains balance after every insertion.

We registered multiple urban zone service nodes into the UrbanAI platform and observed automatic load balancing operations including left-rotation, right-rotation, and double-rotation adjustments. We also implemented a city zone search operation to retrieve infrastructure information efficiently.

Implementation Details:

1. Created the UrbanAI smart city data structure for storing infrastructure service records.
2. Inserted urban zone service records into the platform database.
3. Calculated load balancing factors after every infrastructure update.
4. Performed automatic rebalancing operations whenever service load became unequal.
5. Maintained a balanced city service management structure automatically.
6. Implemented search functionality to locate city infrastructure data quickly.
7. Displayed UrbanAI platform metrics such as zone count, service load, and coverage index.
8. Verified platform optimization property after all zone registrations.

Results and Screenshots:

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MOVIES INSERTED:
Avatar, Inception, Titanic, Joker, Interstellar

AVL INSERTION PROCESS

1) Inserted Avatar
   Root = Avatar

2) Inserted Inception
   Balance Factor of Avatar = -1
   No Rotation Required

3) Inserted Titanic
   Balance Factor of Avatar = -2
   RR Rotation Applied at Avatar

4) Inserted Joker
   Balance Factor of Titanic = 1
   No Rotation Required

5) Inserted Interstellar
   Balance Factor of Titanic = 0
   AVL Property Maintained
```

```
FINAL AVL TREE

      Titanic
     /      \
  Inception  Joker
   /  \      /
Avatar Interstellar

AVL TREE INFORMATION
-----
Root Movie      : Titanic
Tree Height     : 3
Balance Factor Root : 0
AVL Property    : Maintained

SEARCH OPERATION
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Searching for Titanic...

Movie Found

Time Complexity:
Insertion : O(log n)
Search    : O(log n)
```

Time Complexity: UrbanAI Insert, Search, and Delete operations are performed in $O(\log n)$ time due to automatic tree balancing. Compared to an unoptimized city management system, UrbanAI provides significantly better worst-case infrastructure management performance because the service load remains balanced after every zone registration and update.

GitHub Link: <https://github.com/marahrudaydeepak-ctrl/DSA->

Student Signature	Faculty Signature